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SINGAPORE

The Rise: From Swamp to Superpower

PRESENTED BY GROUP 1

An Overview: Singapore's Economic Model

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Solow-Swan Model

Lewis-Dual Sector Model

Human Capital Strategy

Monetary Autonomy

Why Singapore DOLLAR Stays Independent

State Capitalism & Govt. Investment

Land Use and Urban Planning

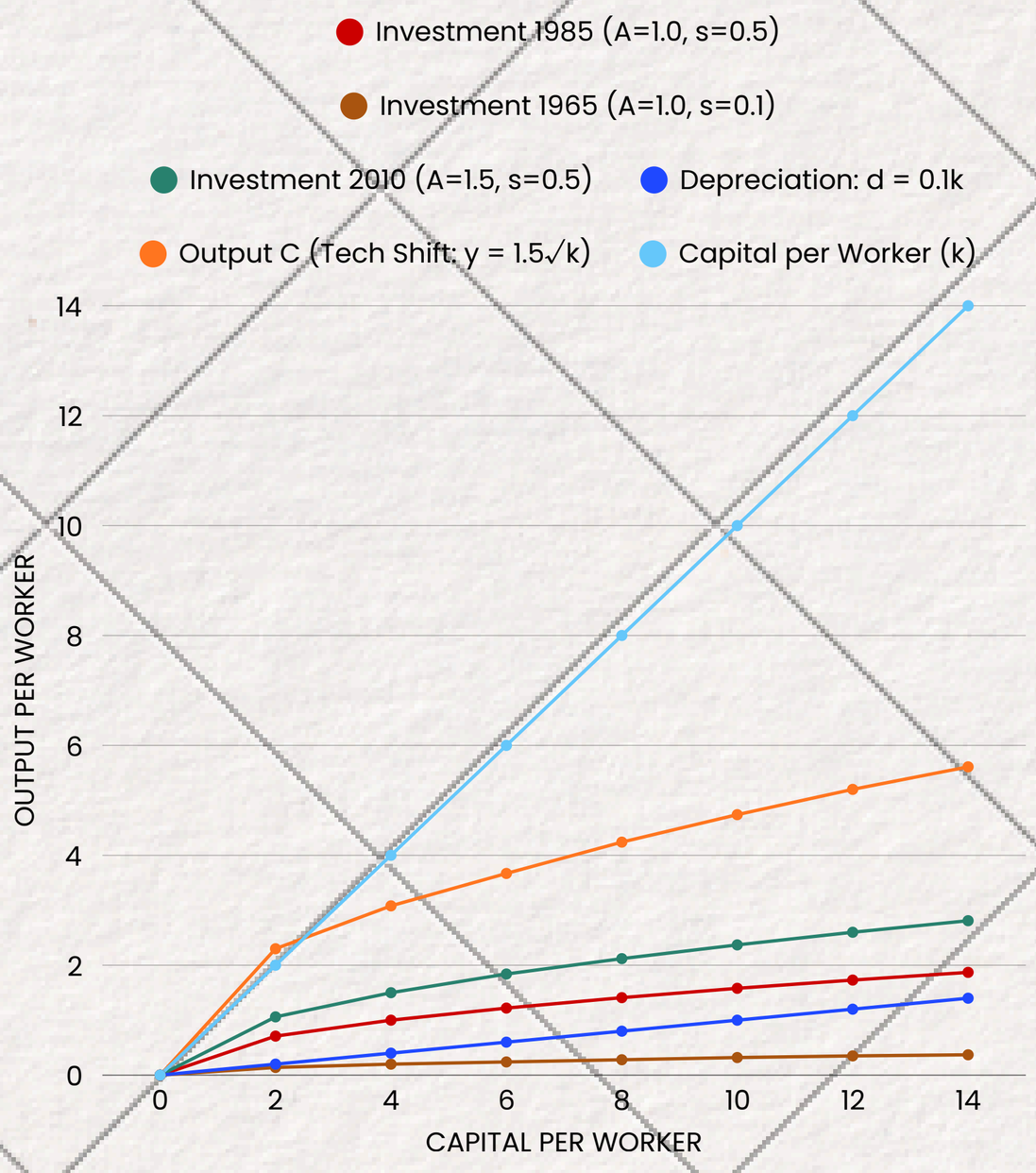
Strong government regulation of the housing market

Governance Law & Behavioral Incentive

School of Thoughts

Conclusion

SOLOW-SWAN MODEL: CAPITAL & TECHNOLOGY ACCUMULATION



Overview

The Starting Formula: Economic growth initially relies on accumulating Capital (K) (machines) and Labor (L); in 1965, Singapore was poor, so the first step was simply gathering these basic building blocks.

Phase 1: "Perspiration": From 1965-1990s, the government used Forced Savings (CPF) to fund rapid infrastructure building, moving the economy along the curve by simply adding more physical machines (K).

The Trap (Diminishing Returns): Eventually, adding more capital stops increasing output (like buying a 3rd laptop for one worker), causing the growth curve to flatten and trapping the economy in the Middle-Income Trap.

Phase 2: "Inspiration" (Tech Shift): To escape stagnation, Singapore shifted focus in the 2000s to Technology (A) and innovation (Fintech/Biotech), improving Total Factor Productivity (TFP)—essentially creating a "better recipe" for growth.

The Escape (Shifting the Curve): Innovation didn't just move Singapore along the old path; it shifted the entire curve upwards (Output C), allowing the economy to grow beyond physical limits by making every unit of capital more efficient.

Entrepot trade- Singapore acts as a middleman (Import good from one country and then export to other).

Singapore uses its strategic Maritime Location to connect major world regions.

Sovereign wealth funds and government linked companies help Singapore stay influential internationally.

Industrialisation- Focus on Labor Intensive Industries. Singapore had no “Industrial Tradition”. Majority of its population was in Trade & Services.

Globalisation enabled Singapore to overcome regional mismatches with its neighbours and successfully build strong trade linkages with Europe and America.

LEWIS DUAL SECTOR MODEL

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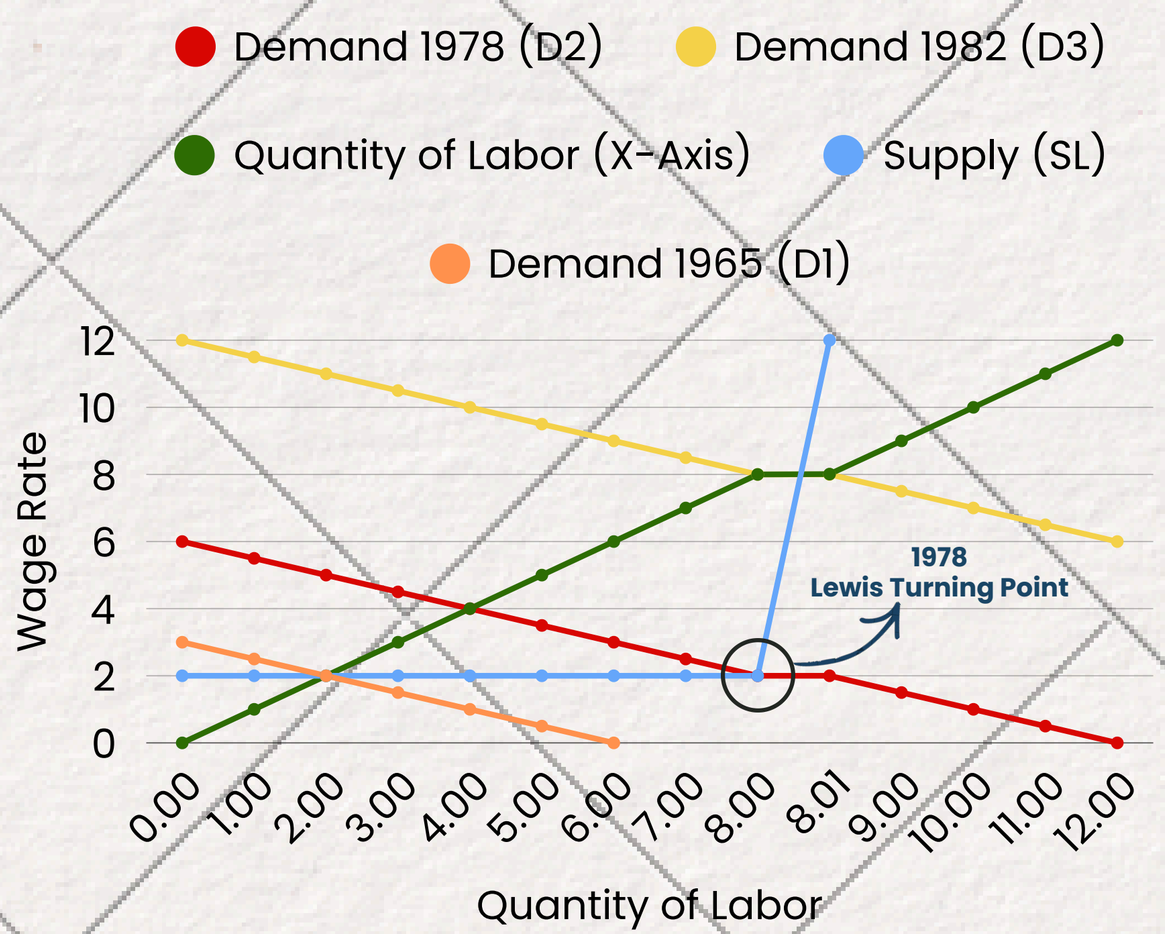
The Lewis Turning Model, developed by **W. Arthur Lewis** in the 1950s, describes the structural transformation of a developing economy from a traditional, labor-surplus agricultural sector to a modern, industrial sector.

It hinges on the "Lewis Turning Point," the moment when surplus rural labor is fully absorbed into manufacturing, causing industrial wages to rise and transitioning the economy towards higher-productivity, capital-intensive growth.



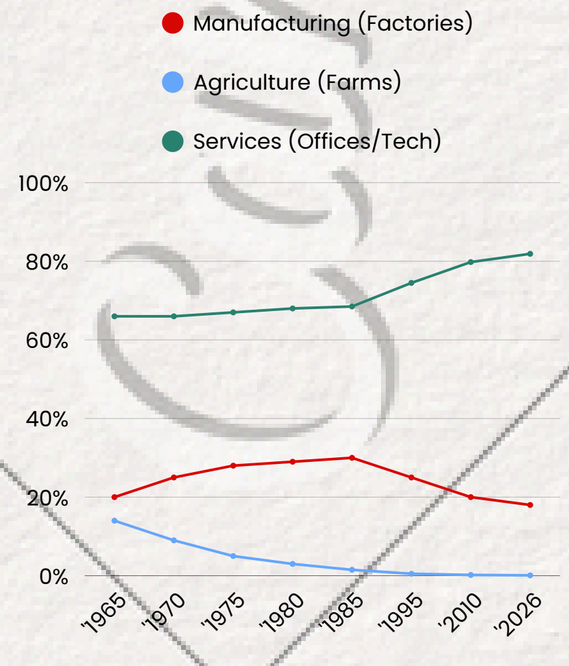
The Unlimited Phase (1965–1978): Singapore began with a massive surplus of labor (a horizontal supply curve), allowing MNCs to hire rapidly at a flat, low wage rate ($2) without triggering inflation or rising costs.

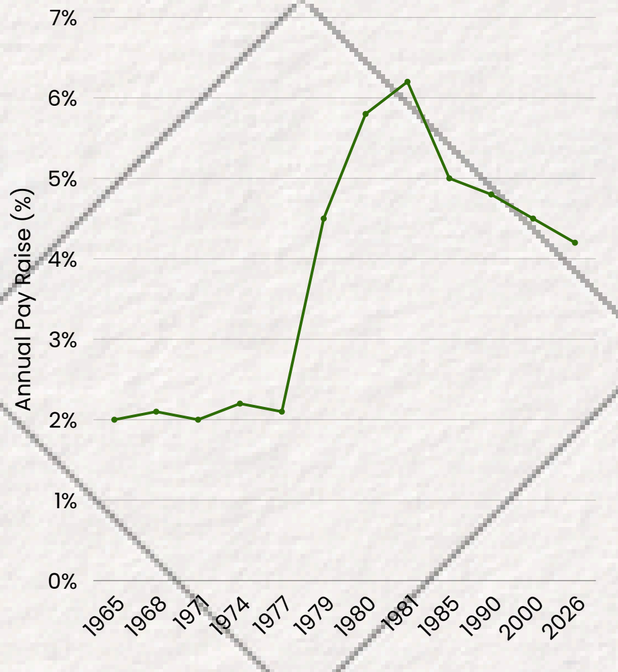
Profit-Led Expansion: As foreign investment shifted labor demand to the right, the economy absorbed workers from traditional sectors (subsistence farming) into manufacturing, fueling high corporate profits and reinvestment.

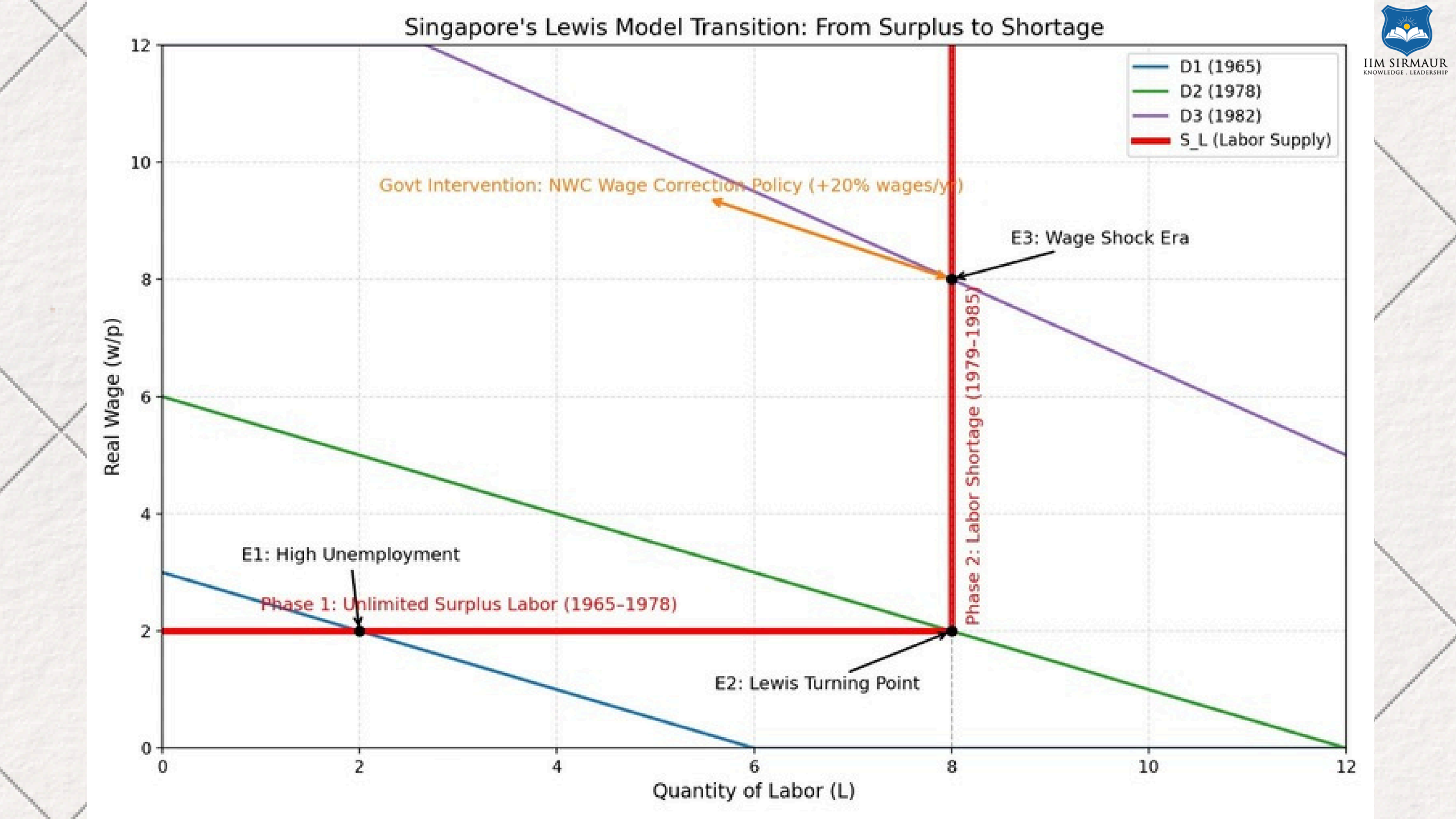
The Lewis Turning Point: By the late 1970s, there were no unemployed labor, causing the labor supply curve to become vertical (inelastic) as the economy hit full employment.

The Wage Shock Intervention: To prevent a return to cheap labor via low-skilled immigration, the government deliberately forced wages up (the Wage Correction Policy), creating an artificial price floor well above market rates.

Forced Upgrading: This wage shock destroyed the profitability of low-value, labor-intensive firms (like textiles), forcing the entire economy to pivot toward high-tech, capital-intensive industries (like electronics) to survive.






Singapore's Lewis Model Transition: From Surplus to Shortage
D1 (1965)
D2 (1978)
D3 (1982)
S_L (Labor Supply)
Govt Intervention: NWC Wage Correction Policy (+20% wages/yr)
E3: Wage Shock Era
Phase 2: Labor Shortage (1979–1985)
E1: High Unemployment
Phase 1: Unlimited Surplus Labor (1965–1978)
E2: Lewis Turning Point
Real Wage (w/p)
Quantity of Labor (L)
0
2
4
6
8
10
12

Singapore's Human Capital Strategy

Transitioning From Factor-Accumulation to Productivity-Driven Growth via Supply-Side Intervention

School of thought:
Endogenous Growth (Romer)
& Human Capital (Becker)

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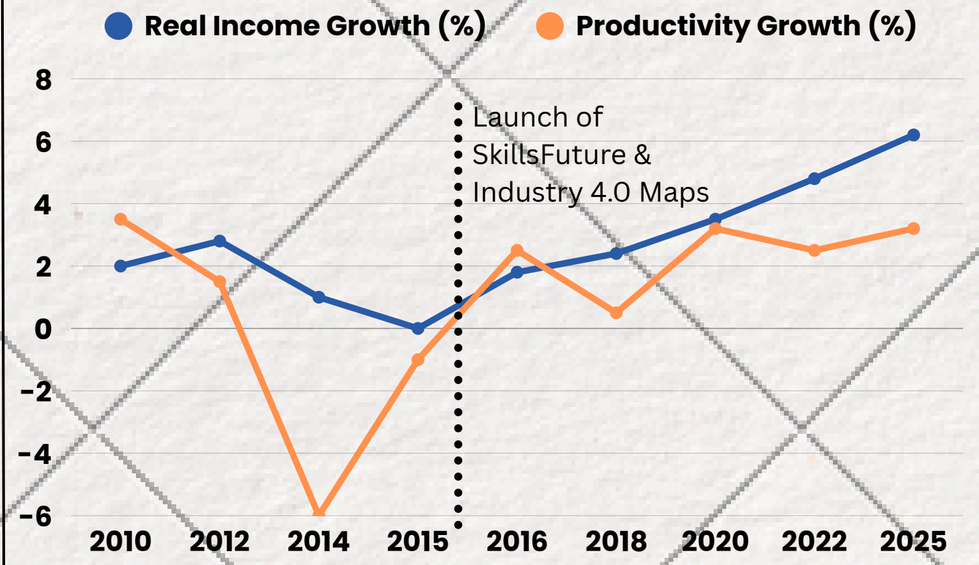
Macro Proof: Visual Evidence

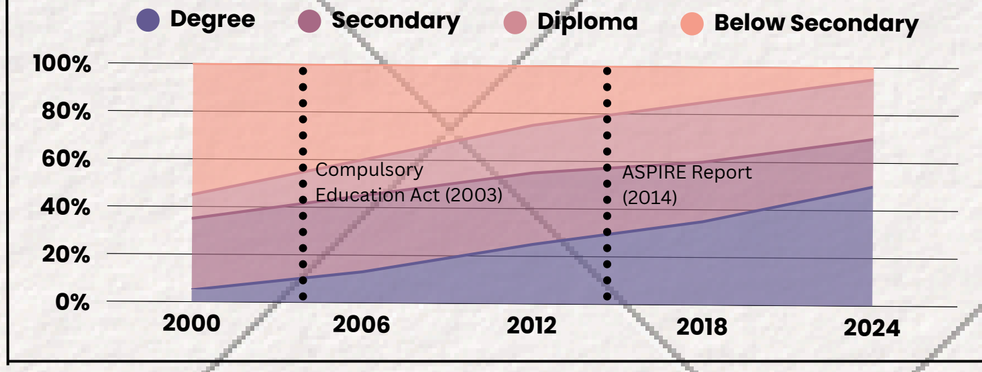
Graph 1: The "Resilience" Curve

(Real Wages VS Labour Productivity Growth 2010-2025)



Graph 2: The "Supply Shock" Shift

(Workforce Compostion by education 2000-2024)



Strategic Mechanisms: Core Analysis

1. Macro Framework (Theory)

- **Endogenous Growth:** Education is treated as an investment in Technology (A), not just Labour (L).
- **Combating Hysteresis:** Lifelong learning prevents "skills decay" as industries pivot (Mfg → Fintech)
- **LRAS Shift:** Upskilling shifts Aggregate Supply rightward, enabling non-inflationary growth.

2. Policy Levers (Action)

- **SkillsFuture (2015):** National credit system decoupling skills from degrees to increase labour mobility.
- **Bilingual Policy:** Lowers transaction costs for FDI; creates "cultural arbitrage" with China/India.

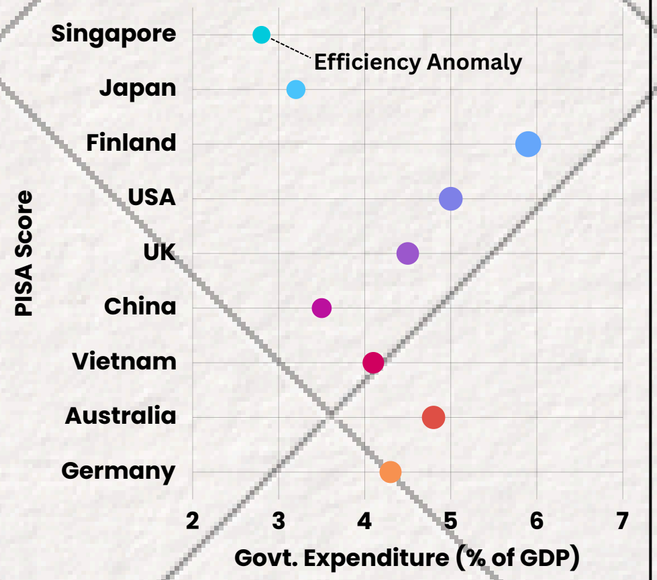
3. Strategic Outcomes (Results)

- **Wage Premium:** Work-Study grads earn **~10% premium** over peers
- **Value Chain:** Tertiary education rose **7% (1994) → >45% (2024)**

The Fiscal Anomaly

Graph 3: The 'Efficiency' Callout

(PISA Score vs. Govt. Spend % GDP)



The Fiscal Anomaly: Top PISA scores at ~50% the cost of OECD peers.

Future Risk and Conclusion

With fertility<1.0, GDP growth relies entirely on increasing Human Capital per capita. Future success depends on solving the "Paper Chase" (Signalling vs. Productivity) issue.

"The only natural resource we have is people" - Lee Kuan Yew

SINGAPORE'S MONETARY AUTONOMY

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Monetary Policy Flexibility

- The key cost of a currency union is the loss of independent monetary policy.
- Policy control allows economies to respond to domestic shocks and stabilize inflation.

For Singapore: Flexibility is essential due to high exposure to global trade cycles.

Asymmetric Shock Risk

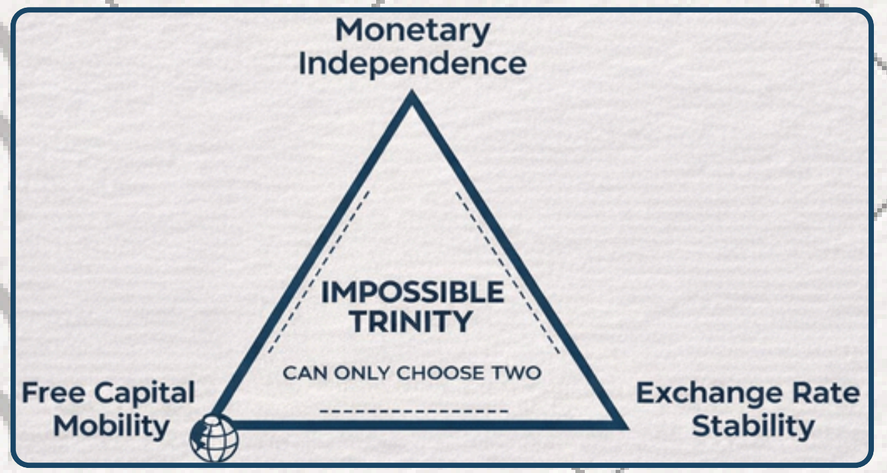
- OCA theory identifies asymmetric shocks as a key factor in determining whether economies should share a currency.

Singapore: As a globally integrated economy, external demand and trade fluctuations can affect it differently: requiring independent policy action.

Small and Highly Open Economy

- Highly open economies often become more specialised, increasing vulnerability to country-specific shocks.

Singapore: Heavy reliance on exports and international trade makes exchange-rate flexibility crucial for macroeconomic stabilization.



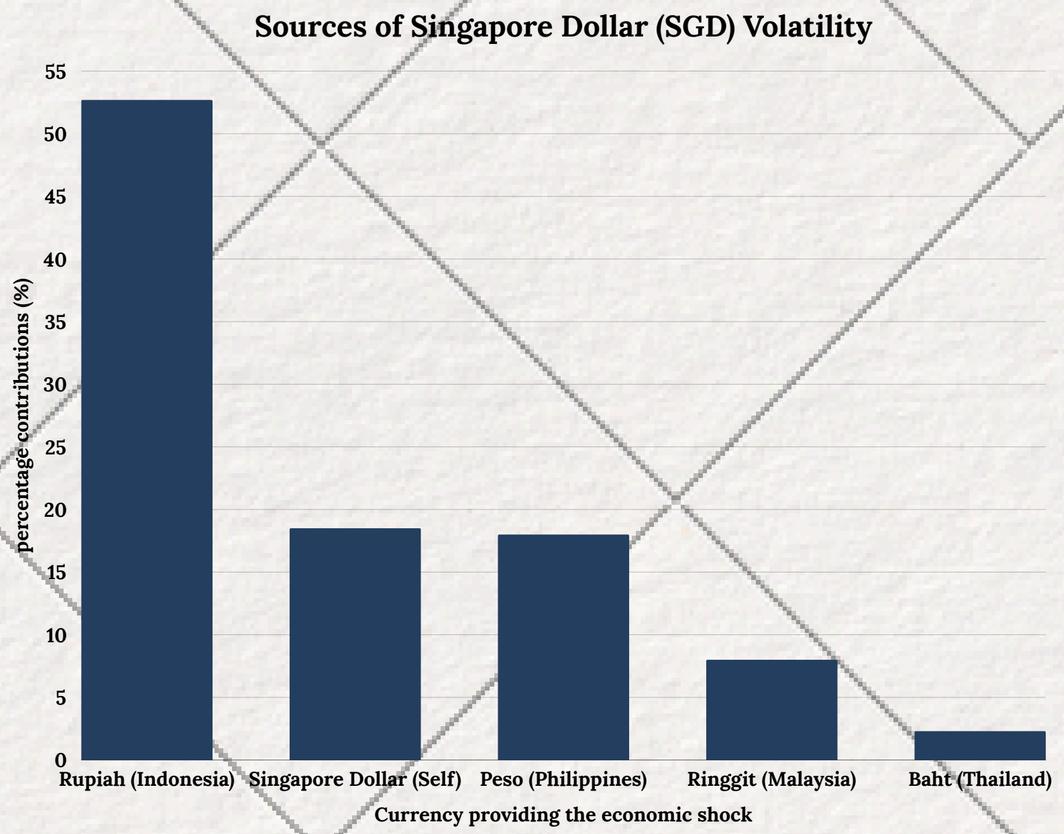
Need for Shock Alignment in a Currency Union

- Countries adopting a single currency should exhibit similar responses to exchange-rate shocks.

Singapore: Its trade-driven structure and relatively stable currency reflect unique macro conditions, meaning a common policy may not suit its economic needs.

- Dr. Hussain Kunroo (2015) Theory of Optimum Currency Areas
- Kim & Chow (2003) Optimum Currency Area in Europe: An Alternative Evaluation
- Riyanto et al. (2021) An Optimum Currency Area in ASEAN 5 Countries

WHY SINGAPORE DOLLAR STAYS INDEPENDENT



ASEAN economies face asymmetric shocks, with differing economic structures and volatility patterns; hence, **a common monetary policy would not suit all members equally.**

External Waves: Over 80% of the currency's movement is triggered by neighbors like Indonesia and the Philippines.

Neighbor Power: 52% of the SGD's value shifts are actually caused by Indonesia, not Singapore.

The Shield: Singapore keeps its own money so it can filter out these outside economic shocks before they hurt local people.

The Verdict: To stay a global trade leader, Singapore must keep its own "steering wheel" to handle waves that hit it differently than its neighbors.

Low regional shock synchronization and limited labour mobility weaken ASEAN's suitability as an Optimum Currency Area.

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STATE CAPITALISM AND GOVERNMENT INVESTMENT

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Singapore's State Capitalism Model

Government ownership combined with market discipline to accelerate industrial growth.

GIC Investment Corp

GIC safeguards national reserves and strengthens long-term financial resilience

Human Capital as Economic Strategy

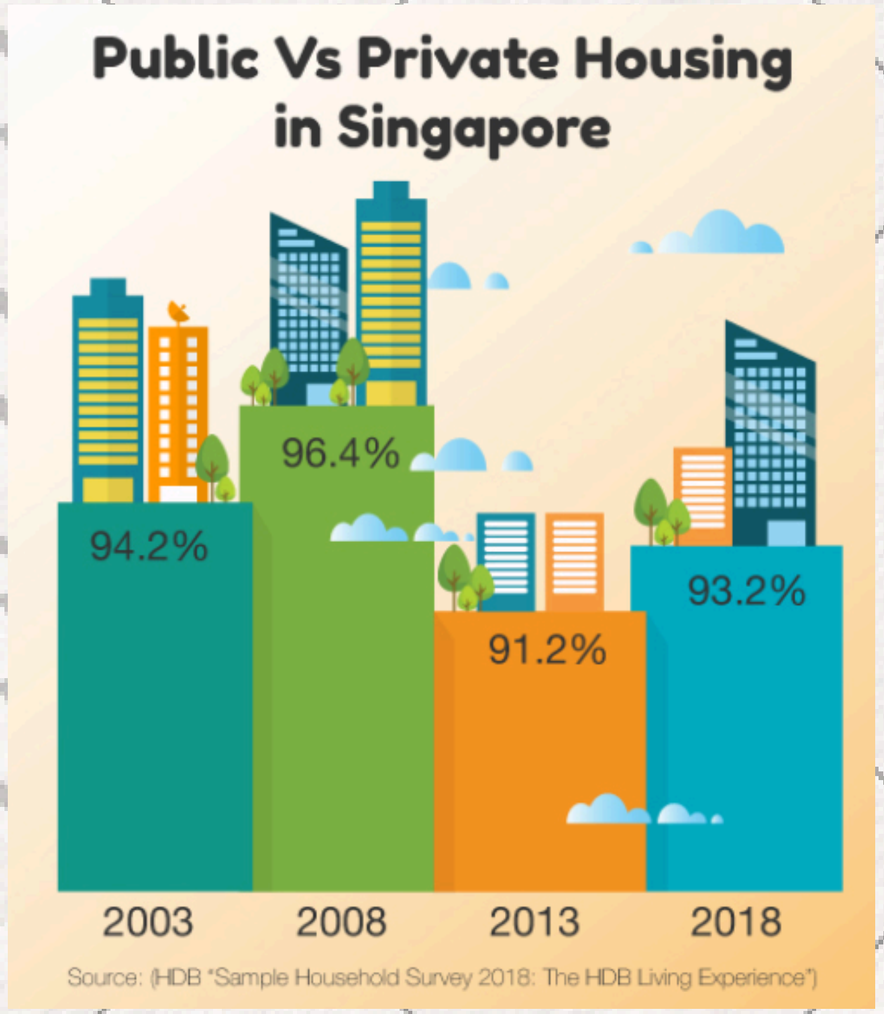
Continuous upskilling policies ensure a future-ready, high-productivity workforce

Strategic State Intervention

Government-linked firms drove early industrialisation where private capital was unwilling to invest.

Temasek Holdings

Temasek actively invests in strategic sectors to sustain national competitiveness.



- Temasek & GIC Portfolios: Temasek Review 2024 & GIC Annual Report 2023/24.
- Land & Housing Data: Singapore Land Authority (SLA) & HDB Annual Report 2024.
- Human Capital: Ministry of Education (MOE) & SkillsFuture Singapore Strategic Research.

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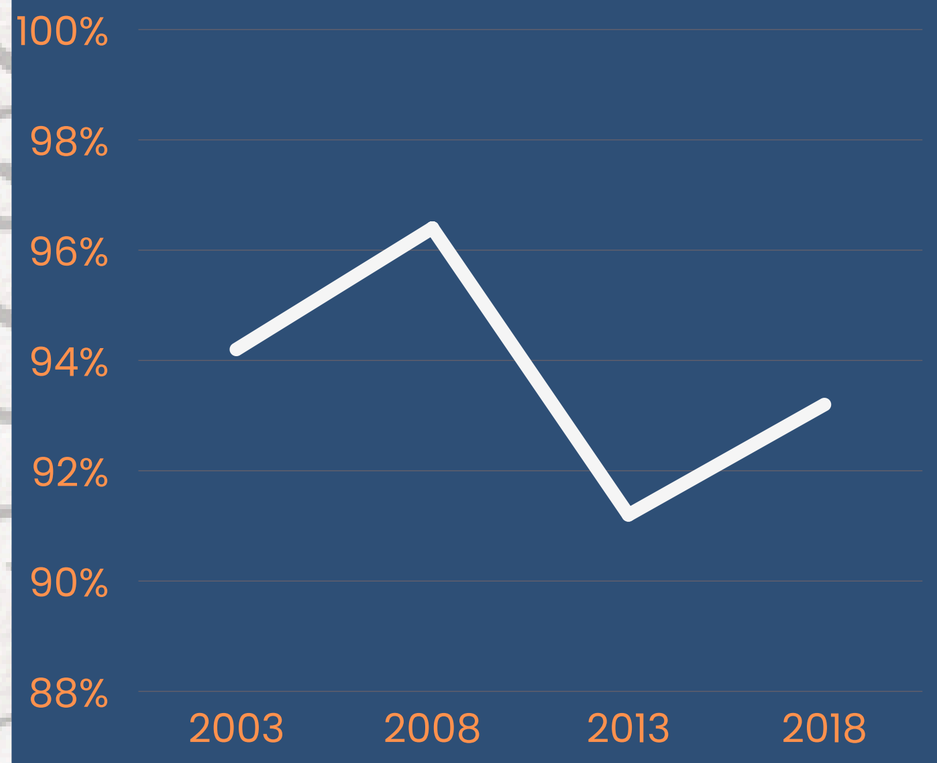
Public Vs Private Housing in Singapore



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STATE-CONTROLLED LAND USE AND URBAN PLANNING

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1) Why the State Controls Land

- The government owns about 90% of Singapore's land.
- Most land is sold on 99-year leases, not freehold.
- This allows the state to control what is built, where, and when.

2) Role of the Urban Redevelopment Authority (URA)

Planning is managed by the Urban Redevelopment Authority (URA).
URA creates:

- Concept Plan (long-term 40–50 years)
- Master Plan (reviewed every 5 years)

They decide:

- Residential zones
- Business districts
- Industrial areas
- Green spaces

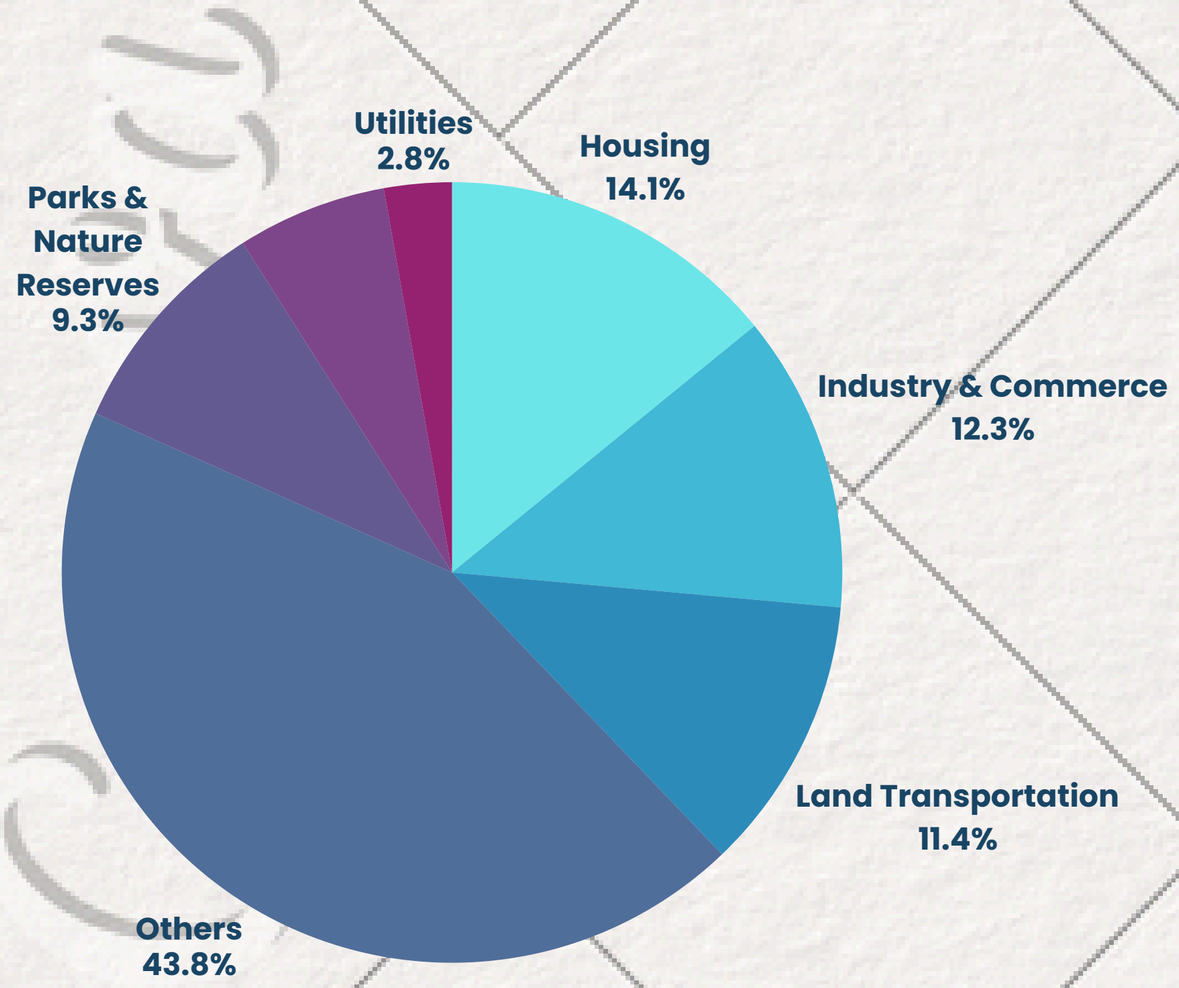
3) Decentralisation Strategy

Instead of one crowded city center:

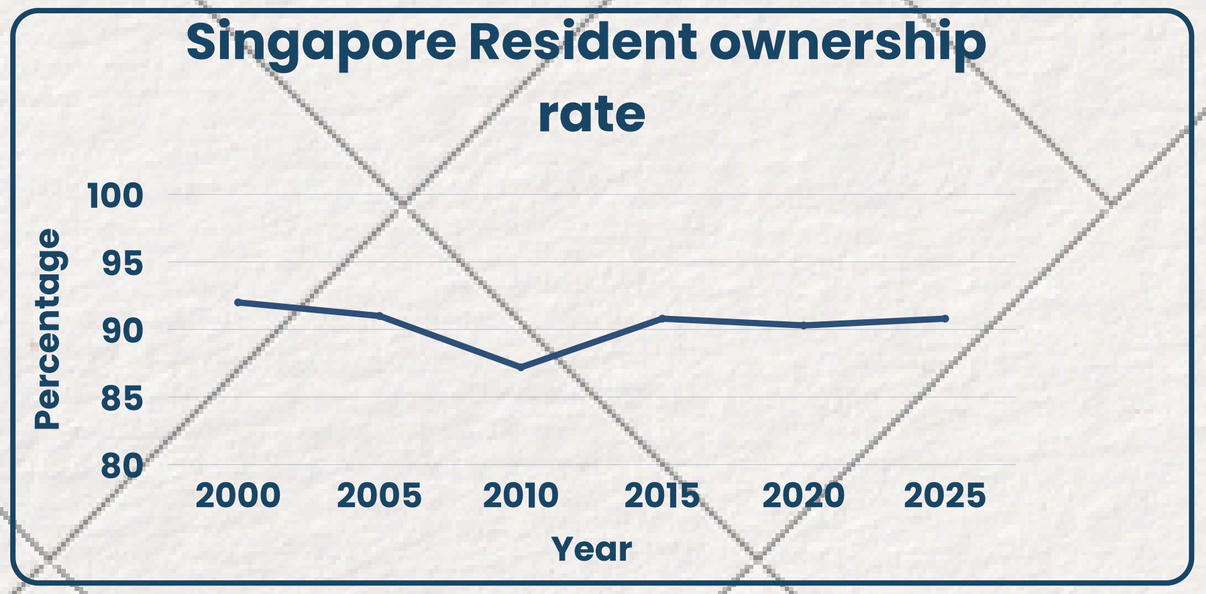
- Singapore created regional business hubs like Tampines and Jurong.
- This spreads jobs across the island.

Result:

- Less traffic congestion
- More balanced economic growth



STRONG GOVERMENT REGULATION OF THE HOUSING MARKET

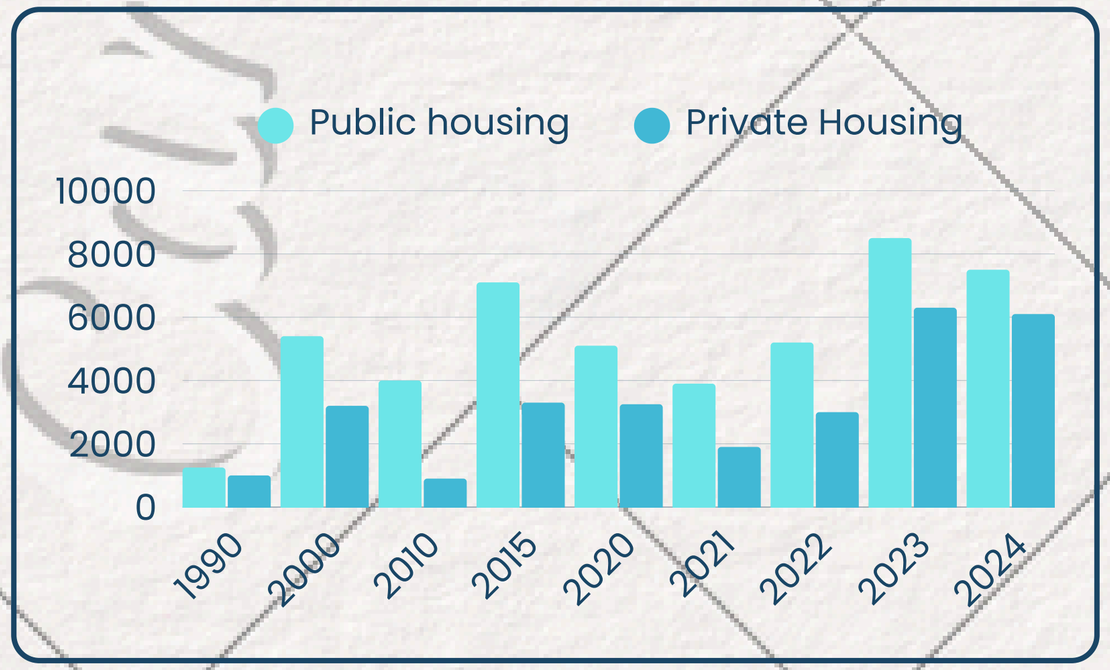


1. Strategic DNA

Stability: Eliminates speculative bubbles and financial contagion.
Equity: Prioritizes lifelong affordability over raw market profit.
Social: Anchors the population through tangible stakeholding.

2. State-Led Supply

Monopoly: State ownership of ~90% of land dictates market rhythm.
The BTO Pivot: Just-in-time construction prevents "ghost towns" or supply gluts.
Pricing Power: Government control of land costs decouples housing from runaway inflation.



3. Demand-Side Controls: Preventing a Hong Kong Scenario

Unlike Hong Kong, where supply constraints led to severe unaffordability and about 2.7% of residents live in “coffin homes,” Singapore’s HDB provides subsidised housing to lower-middle-income groups.
Broad public access and large-scale supply prevented extreme inequality and chronic shortages.

4. Financial Discipline: Preventing Housing Bubbles

Singapore also guards against speculative excess. Mortgage repayments are capped at roughly one-third of household income, and total debt obligations are limited to about 55%.
These safeguards keep prices aligned with wages and reduce systemic risk.

GOVERNANCE, LAW AND BEHAVIOURAL INCENTIVE

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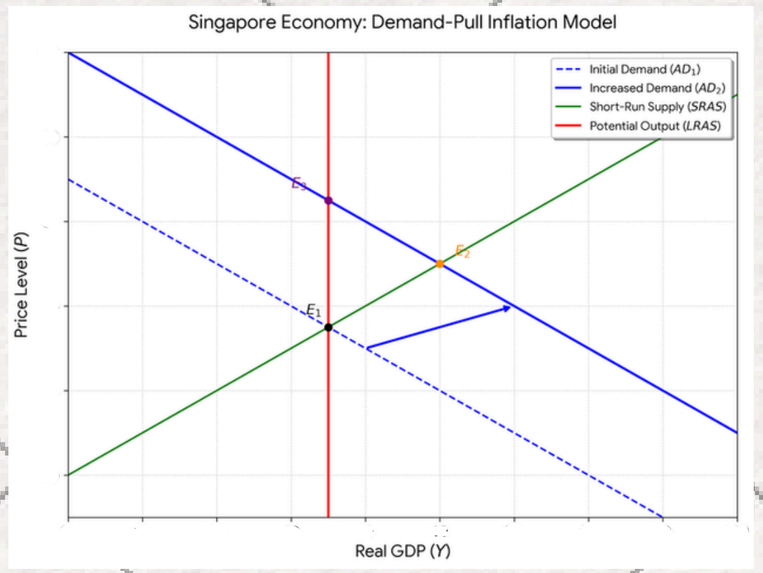
Governance & The Production Function

- $Y=A \cdot f(K,L)$
- By establishing a meritocratic bureaucracy and zero tolerance for corruption, Singapore increased its efficiency. The same amount of capital (K) and labor (L) produced more output (Y) because there was less "leakage" through bribery or mismanagement.
- Effective governance shifted the Long-Run Aggregate Supply curve to the right. While other nations faced stagflation in the 1970s, Singapore's stable governance ensured that supply kept pace with demand.
- The government heavily invested (G) in vocational training and education to ensure the labor force (L) could handle increasingly complex technology (K).



- This ensured that for every unit of input, the output (Y) was maximized, preventing supply slow down even as demand grew.

Law & The Investment (I) Component of GDP

- $Y=C+I+G+(X-M)$
- Institutions are weak, the marginal cost of investment is high.
- By adopting a rigorous, English-style Common Law system, Singapore provided Legal Certainty.

Economic Expansion Incentives Act (1967)

- It granted pioneer status to foreign firms, offering tax exemption (fiscal policy) for up to five years.
- By significantly lowering the marginal cost of investing in Singapore, it triggered a massive inflow of Foreign Direct Investment (FDI), helping the economy transition from labor-intensive to capital-intensive.

Employment Act (1968)

- Main labour law, providing basic terms and working conditions for most employees, including foreigners and local staff. It also standardized working conditions and restricted the power of unions to strike, essentially mandating industrial peace.
- It stabilized the labor market, ensuring that government infrastructure spending (G) successfully "crowded in" private FDI by making the production environment predictable.

GOVERNANCE, LAW AND BEHAVIOURAL INCENTIVE

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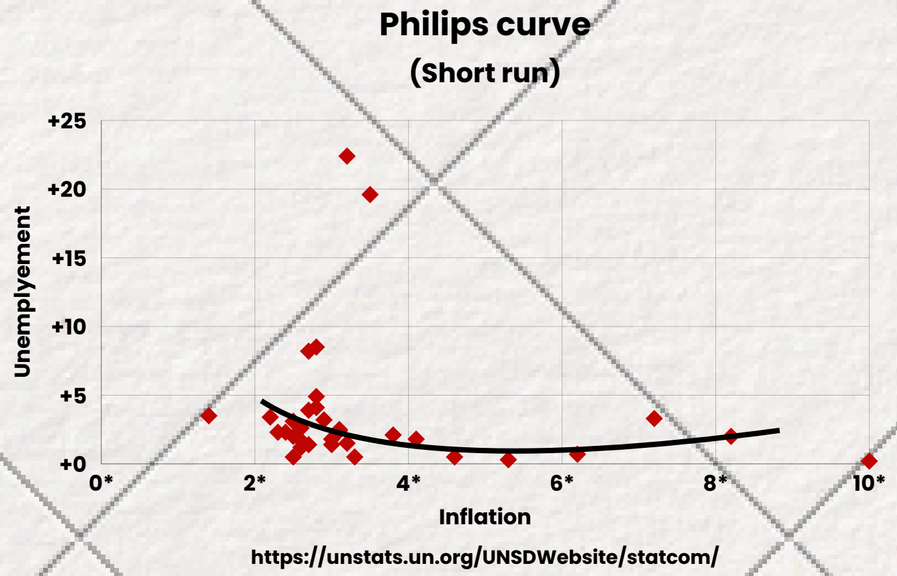
How Singapore's Behavioral System Affects the Phillips Curve



Feature	1965–1997	1965–2024
Time period	Early development & industrializatio	Mature advanced economy
Unemployment range	Very wide (≈10% → 1–3%)	Narrow (mostly 2–3%)
Inflation range	Very wide (–2% → 22%)	Moderate (–1% → 6%)
Trade-off visibility	Clear negative relation	Weak relation
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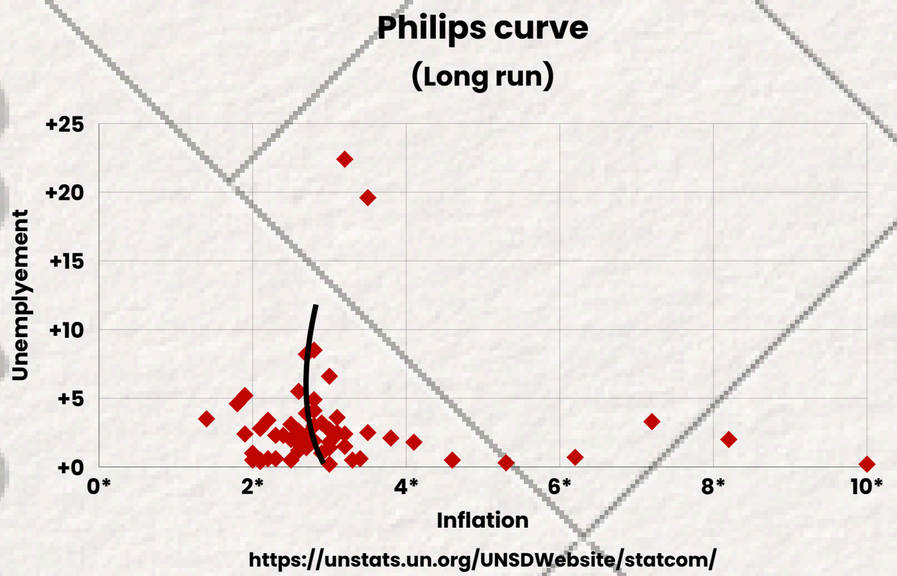


"Behavioral incentives in Singapore anchor inflation expectations and wage behavior, flattening the Phillips Curve and allowing lower unemployment without accelerating inflation."

Anchored Inflation Expectations

Strong governance + policy credibility means workers and firms believe inflation will stay low.

- No panic wage demands
- No defensive price hikes

Result: The Phillips Curve becomes flatter Lower inflation response for any given fall in unemployment.

Disciplined Wage Behavior

Merit-based pay, weak populist pressure, and flexible labor markets reduce wage rigidity.

- Fewer strikes
- Performance-linked wages
- Acceptance of macro constraints

Result: The Phillips Curve becomes flatter Lower inflation response for any given fall in unemployment.

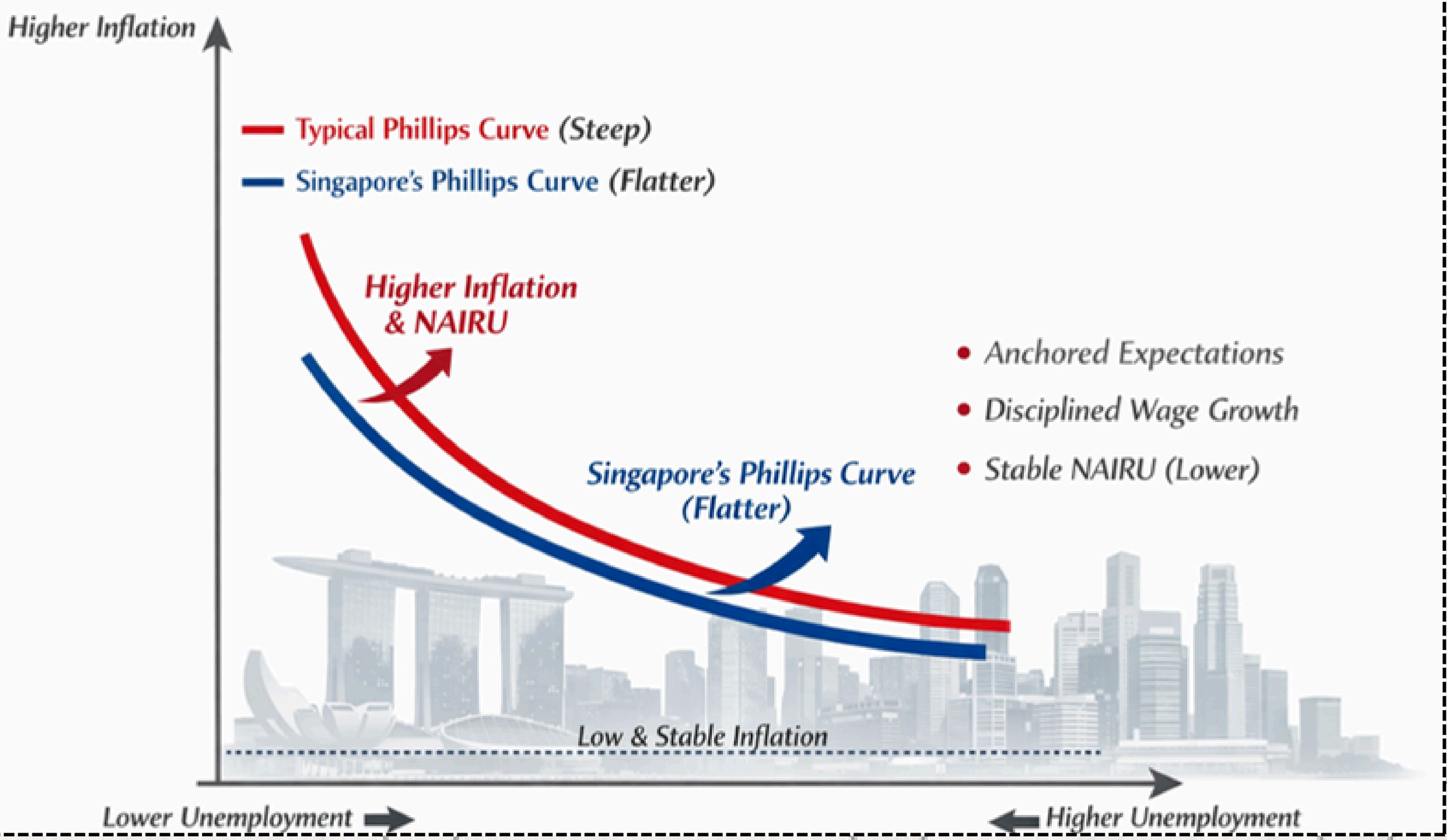
Reduced Cost-Push Inflation

Low corruption and high compliance mean:

- Fewer supply bottlenecks
- Lower informal costs (bribes, delays)

Result: Short-run Phillips Curve shifts downward Inflation is lower at every unemployment level.

The Phillips Curve in Singapore



SCHOOL OF THOUGHTS

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Classical

Singapore's free trade and openness fit classical ideas well. But strong state control and government-linked firms go beyond a pure market system.

Neo-Classical

High savings, human capital, and productivity explain Singapore's growth. Still, the government played a larger role than theory typically assumes.

Cambridge Keynesian

State-led industrial policy clearly helped Singapore transform. But its heavy reliance on global firms adds a strong market element.

Keynesian

"Public spending on housing, education, and infrastructure supported stability and growth. Yet fiscal policy remained stricter than classic Keynesian models."

Monetarist

Low inflation and stable policy built strong investor confidence. However, the level of state intervention is higher than monetarists prefer.

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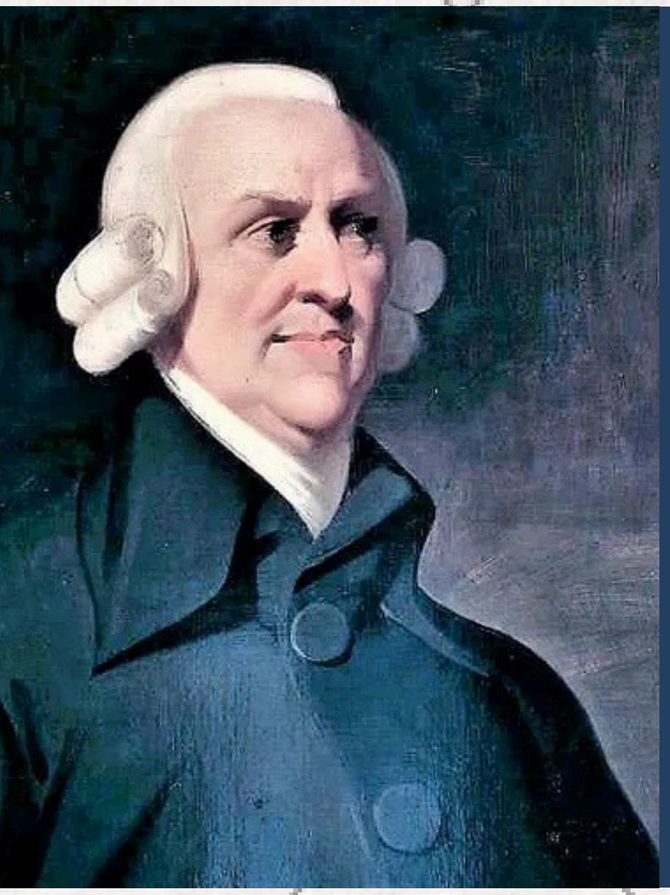

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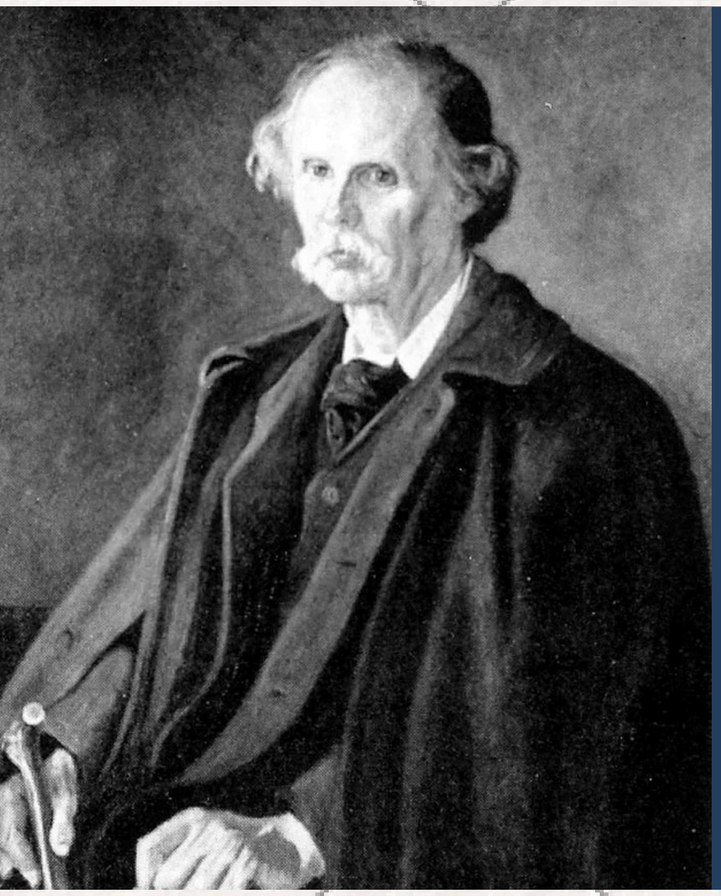

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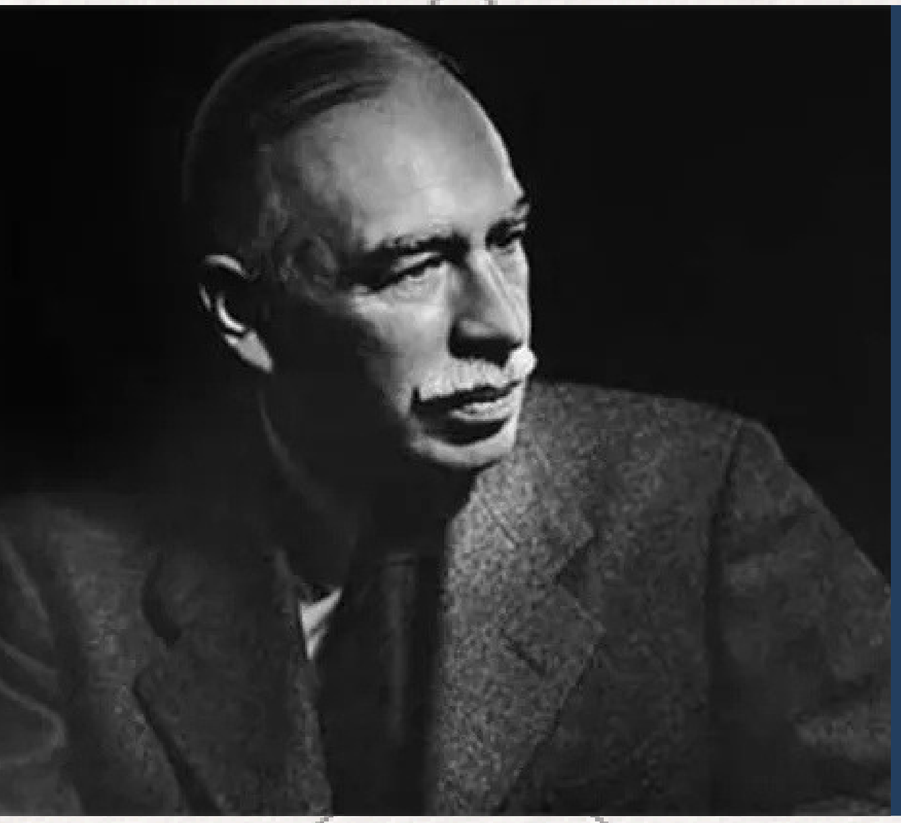

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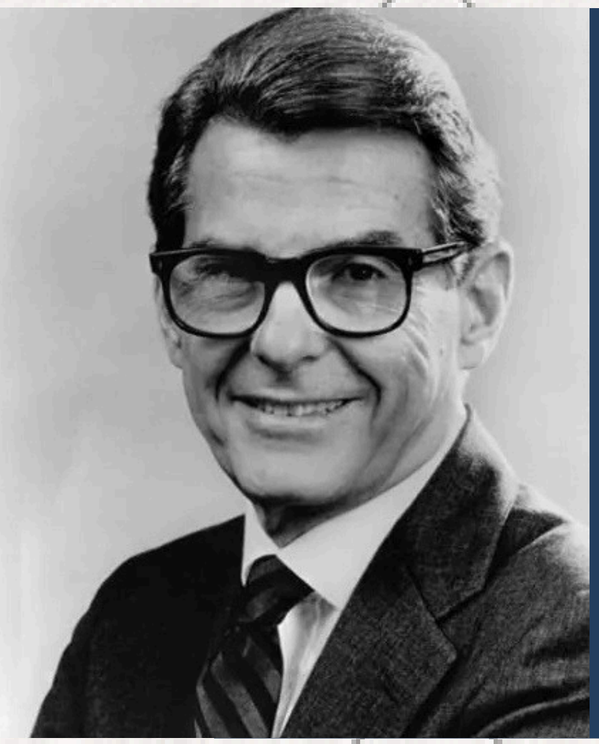

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State-led industrial policy clearly helped Singapore transform. But its heavy reliance on global firms adds a strong market element.

Keynesian

"Public spending on housing, education, and infrastructure supported stability and growth. Yet fiscal policy remained stricter than classic Keynesian models."

Monetarist

Low inflation and stable policy built strong investor confidence. However, the level of state intervention is higher than monetarists prefer.

SCHOOL OF THOUGHTS

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Monetarist

Low inflation and stable policy built strong investor confidence. However, the level of state intervention is higher than monetarists prefer.

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